



Bridge Inspection Report

180430C00812001
COLDWATER CRK LANE
over
COTTONWOOD CREEK TRIB



Inspection Date: 03/18/2023

Inspected By: Modjeski and Masters, Inc.

Anthony Schoenecker, PE (Team Leader)
Michael Parr, PE

Inspection Type(s): Routine



5/16/2023

Modjeski and Masters, Inc. prepared for Srinteg Corporation | Contract No. 88-2IDP5044 | Work Authorization No. 1

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SECURITY ACT AND 23 USC SECTION 409, SAFETY SENSITIVE INFORMATION

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Latitude: 33.05138

Longitude: -96.651132

Bridge Inspection Follow-Up Action

Initial Finding Information

FUA ID 658796

Location Information

Bridge ID# 180430C00812001 Latitude 33.05138 Longitude -96.651132 County 043-COLLIN Location .05 MI E OF PARKER RD

Facility Carried COLDWATER CRK LANE Feature Crossed COTTONWOOD CREEK TRIB

Ownership/Responsibility

District DAL Area Office Maintenance Section 07 Control - Section C008 - 12
System Designation 2 Bridge Owner City of Plano Precinct

Maintenance Information

Inspection Date 03/18/2023 Rec Type Level 3 Previously Recommended? Y Critical Finding? N

Bridge Component Channel Repair Category Repair Bank Erosion/Provide Erosion Countermeasures

Other:

Description of Issue and Recommendation

A large area of erosion is present behind the northeast wingwall up to 2' deep, and scour has exposed the wingwall footing and upstream apron toewall up to 1.5' and is beginning to undermine at the northeast corner. There is erosion up to 1' deep behind the northwest wingwall.

Recommendation: Fill voids behind the northeast and northwest wingwalls and provide scour countermeasures.

District Bridge Office Comments

Comments By Date

District Area or Assigned Maintenance Office Comments

Comments By Date

Follow-up Action Resolution Comments

Work to be Completed Via: +Project Status: +District Estimated Letting Date:
+District Proposed Funding Source: +CSJ / WO / Project ID: +Optional Fields

Action Taken: Verified By Date

+Additional Comments if Necessary:

CRITICAL FINDING FOLLOW UP INFORMATION (BLANK IF FOLLOW UP ACTION IS NOT A CRITICAL FINDING)

Condition Rating (Affected elements only) Overall Recommendation: New Load Rating / Analysis Required?

Follow Up Inspection Information:

Automated Email Sent To:
Date Bridge Division notified of Critical Finding: Date Bridge Owner notified of Critical Finding: Concur with findings:
TxDOT Inspector: Co-Inspection Date w/ Bridge Owner: Bridge Owner Rep:
Comments:

Temporary Repair Information:

TxDOT Inspector: Date Temporary Recommended Repair(s) Discussed w/Bridge Owner:
Bridge Owner Rep: Co-Inspection Date w/Bridge Owner:
Temp Repair In Place: Date Temporary Repair was Installed: Date Bridge Division notified of Temp Repair:
Comments:

Permanent Repair Information:

TxDOT Inspector: Permanent Repair Target Date:
Bridge Owner Rep: Co-Inspection Date w/Bridge Owner:
Date Permanent Recommended Repair(s) Discussed w/Bridge Owner: Permanent Repair Construction Date: Date Bridge Division notified of Permanent Repair:
Comments:

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Inspector: Anthony Schoenecker

Structure Number:

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Facility Carried:

Bridge Inspection Report

Pictures



PHOTO 1 Recommended Maintenance Needs

Description Large area of erosion behind NE wingwall (up to ~ 2' deep), footing at end of wingwall and upstream apron toewall are exposed up to 1.5' and beginning to undermine, looking south.



LOAD POSTING INFORMATION

Structure Number
Facility Carried
Feature Intersected

180430C00812001
COLDWATER CRK LANE
COTTONWOOD CREEK TRIB

Inspector
Inspection Date
Maint. Section

Schoenecker,Anthony
03/18/2023
07

Components & Ratings:

	Component Rating	Inventory LR		Operating LR	
		H	HS	H	HS
Deck (Item 58)	N				
Superstructure (Item 59)	N				
Substructure (Item 60)	N				
Culvert (Item 62)	7		20		33.4

☐ Posting Requirement Exception: The governing condition rating is controlled by a non-load path element. Therefore, load posting is not recommended.

Seal:

Signature & Date:

Prime Firm Name & Number:

Sub Firm Name & Number:

Load Posting Limits:

	Sign Type	Limit (lbs)					
		Gross	Single Axle	Tandem Axle	Axle or Tandem	Comb. Vehicle	Single Unit Veh. X + Axles
Current Rec. ()							
Prev. Recommended							
Observed Posting							

Material Needed:

Sign Type

Quantity

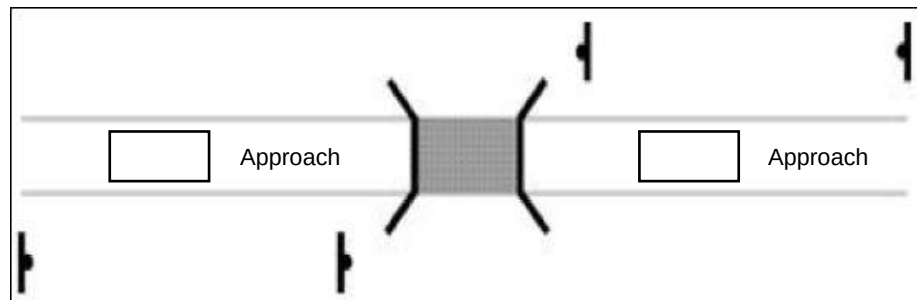
Sign Type

Quantity

Posts

Hardware Sets

Decals



Approach Roadway: Sign Ty. Condition Code Maintenance Need
 Advance Warning
 Advance Warning
 Approach

Approach Roadway: Sign Ty. Condition Code Maintenance Need
 Advance Warning
 Advance Warning
 Approach

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BRIDGE INSPECTION RECORD

Structure Number
Facility Carried
Feature Intersected

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COLDWATER CRK LANE
COTTONWOOD CREEK TRIB

Inspector
Inspection Date
Maint. Section

Schoenecker,Anthony
03/18/2023
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Condition Rating	Description*
N	Not Applicable
9	Excellent Condition
8	Very Good Condition - no problems noted
7	Good Condition - some minor problems
6	Satisfactory Condition - minor deterioration of structural elements (limited)
5	Fair Condition - minor deterioration of structural elements (extensive)
4	Poor Condition - deterioration significantly affects structural capacity
3	Serious Condition - deterioration seriously affects structural capacity, local failures possible
2	Critical Condition - advance deterioration of primary elements, bridge should be closed until repaired
1	Failing Condition - bridge closed but repairable
0	Failed Condition - bridge closed and beyond repair

*These descriptions are for items 58, 59, 60 and 65. Code Items 61 and 62 according to the TxDOT Coding Guide.

Bridge Description:

3 - 8' x 8' x 79.3' Concrete Multiple Box Culvert at 30° RFS

Seal:

Signature & Date:

Prime Firm Name & Number:

Sub Firm Name & Number:

General Comments:

3 - 8' x 8' x 79.3' Concrete MBC
Inventory Direction: West to East

DECK (ITEM 58)

Structure ID: 180430C00812001 Inspection Date: 03/18/2023

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Component	Description	Min.	Rating	Comment
Deck - Component Rating		1	N	
Wearing Surface	32" Concrete and Fill	6	7	Transverse and longitudinal cracks are common along the concrete pavement.
Surfacing Thickness:	8 inches			
Fill Height:	24 inches			
Joints, Expansion, Open		6	N	
Joints, Expansion, Sealed		6	N	
Joints, Other		6	N	
Drainage System		6	8	
Curbs, Sidewalk & Parapets		6	8	
Median Barrier		6	N	
Railings	Non-standard Steel @ South	6	7	Light surface rusting at isolated paint system failures.
Railing Type:				
Railing Protective Coating		7	7	Light surface rusting at isolated paint system failures.
Delineation		7	N	
Other		-	N	

SUPERSTRUCTURE (ITEM 59)

Structure ID: 180430C00812001 Inspection Date: 03/18/2023

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Component	Description	Min.	Rating	Comment
Main Members - Steel		0	N	
Main Members - Concrete		0	N	
Main Members - Timber		0	N	
Main Members - Connections		0	N	
Floor System Members		1	N	

Floor System Connections	1	N
Secondary Members	5	N
Secondary Mem. Connections	5	N
Expansion Bearings	6	N
Fixed Bearings	6	N
Steel Protective Coating	6	N
Other	-	N

Overall Component Rating	N
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SUBSTRUCTURE (ITEM 60)

Structure ID: 180430C00812001 Inspection Date: 03/18/2023
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Component	Description	Min.	Rating	Comment
Abutment Caps		0	N	
Above Ground		0	N	
Below Ground or Foundation		0	N	
Backwalls & Wingwalls		0	N	
Intermediate Supports:				
Caps - Concrete		-	N	
Caps - Steel		-	N	
Caps - Timber		-	N	
Above Ground - Concrete		-	N	
Above Ground - Steel		-	N	
Above Ground - Timber		-	N	
Above Ground - Masonry		-	N	
Below Ground or Foundation		-	N	
Collision Protection System		5	N	
Steel Protective Coating		6	N	

Overall Component Rating	N
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CHANNEL (ITEM 61)

Structure ID: 180430C00812001 Inspection Date: 03/18/2023
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Component	Description	Min.	Rating	Comment
Channel Banks		0	6	A large area of erosion is present behind the northeast wingwall up to 2' deep. Scour has exposed the wingwall footing and upstream apron toewall up to 1.5' and is beginning to undermine at the northeast corner. Minor to moderate erosion along other channel banks.
Channel Bed		0	7	Scour has exposed the wingwall footing and upstream apron toewall up to 1.5' and is beginning to undermine at the northeast corner.
Rip Rap, Toe Walls & Apron		5	7	Scour has exposed the wingwall footing and upstream apron toewall up to 1.5' and is beginning to undermine at the northeast corner.
Dikes		5	N	
Jetties		5	N	
Other		-	N	

Overall Component Rating	6
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CULVERTS (ITEM 62)

Structure ID: 180430C00812001 Inspection Date: 03/18/2023
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Component	Description	Min.	Rating	Comment
Top Slabs		0	7	Transverse cracks (< 0.012" wide) with efflorescence in underside of top slabs. Isolated shallow spalls (< 4" dia.) with exposed reinforcing at the upstream ends of each barrels.
Bottom Slab or Footing		0	8	
Abutments, Intermed. Supports		0	7	Vertical and horizontal cracks (up to 0.012" wide) in abutment and interior walls.
Headwalls & Wingwalls		5	7	Vertical cracks (< 0.012" wide) with isolated light efflorescence. Minor spalls of headwalls at connection to top slab. Vertical cracks (< 0.012" wide) in wingwalls at connections to abutments. Diagonal cracks in southeast wingwall.
Other		-	N	

Overall Component Rating	7
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APPROACHES (ITEM 65)

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Component	Description	Min.	Rating	Comment
Embankments		0	6	A large area of erosion is present behind the northeast wingwall up to 2' deep. There is erosion up to 1' deep behind the northwest wingwall. Erosion and settlement of the embankment between sidewalk and curb at the northeast corner connection to culvert, sidewalk has slightly settled. Erosion (up to 6" deep) is slightly undermining sidewalk near southwest corner of culvert.
Embankment Retaining Walls		4	N	
Slope Protection		5	6	A large area of erosion is present behind the northeast wingwall up to 2' deep. There is erosion up to 1' deep behind the northwest wingwall. Erosion and settlement of the embankment between sidewalk and curb at the northeast corner connection to culvert, sidewalk has slightly settled. Erosion (up to 6" deep) is slightly undermining sidewalk near southwest corner of culvert.
Roadway	Concrete	5	7	A previously deteriorated area of the east approach concrete pavement, next to a large concrete patch near connection to culvert, has been repaired with an asphalt overlay. Transverse and longitudinal cracks are common along the approach concrete pavement.
Relief Joints		6	N	
Drainage		6	7	Settlement of east approach roadway.
Guardfence		6	N	
Delineation		7	N	
Sight Distance		7	8	
Other		-	N	
Overall Component Rating			6	

MISCELLANEOUS

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Component	Description	Min.	Rating	Comment
Signs		N	N	
Illumination		N	N	
Warning Devices		N	N	
Utility Lines		7	N	
Other		N	N	

APPRAISAL RATINGS

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TRAFFIC SAFETY FEATURES (ITEM 36)

	Rating	Comment
0 - Does not meet current standards	0	No Railing or Guardfence provided.
1 - Meets current standards	0	No Railing or Guardfence provided.
N - Not applicable	0	No Railing or Guardfence provided.
Appr. Rails:	0	No Railing or Guardfence provided.
Appr. Rail Ends:	0	No Railing or Guardfence provided.

WATERWAY ADEQUACY (ITEM 71)

7 Est. Deck Overtopping Freq: 11-100 yrs.
Est. Approach Overtopping Freq: 11-100 yrs.

APPROACH ROADWAY ALIGNMENT (ITEM 72)

8 Regulatory/Advisory Speeds: 30 mph/None



BRIDGE INVENTORY RECORD

Structure Number	180430C00812001	Inspector	Schoenecker,Anthony
Facility Carried	COLDWATER CRK LANE	Inspection Date	03/18/2023
Feature Intersected	COTTONWOOD CREEK TRIB	Maint. Section	07

District	18-Dallas	County	043-COLLIN	Latitude	33.05138
Milepoint	12.001	System	2 - Off System	Longitude	-96.651132
General Description: 3 - 8' x 8' x 79.3' Concrete Multiple Box Culvert at 30° RFS					
Bridge Length (ft.)	31	Deck Width (ft.)	0.0		
Skew Angle (degrees)	30	Clear Width Between (ft.)	0.0		
Right / Left Forward	Right Forward	Measurement Feature			
Vertical Over-Clearance	99.99	☒ Unimpaired Over-Clearance			
Surfacing Type	32" Concrete and Fill	Deck Type	See entry for 107.1		
Lanes On	02	Lanes Under	00		
Bridge Rail	None	Approach Roadway Width (ft.)	36.0		

Seal:

Signature & Date:

Prime Firm Name & Number:

Sub Firm Name & Number:

Superstructure:	Spans		
Main:	Type	Size	Number
	Spacing	Controlling Span Length (C-C bearings)	
Superstructure:	Spans		
Major Approach:	Type	Size	Number
	Spacing	Controlling Span Length (C-C bearings)	
Superstructure:	Spans		
Minor Approach:	Type	Size	Number
	Spacing	Controlling Span Length (C-C bearings)	

Estimated Deck Overtopping Frequency	11 - 100	Comments:
Estimated Approach Overtopping Frequency	11 - 100	
Horizontal / Vertical Alignment	Good / Good	
Regulatory / Advisory Speeds	30 mph / None	
Year Built	2000	Year(s) Widened
Posted Load Restriction	NNN	☒ No Restriction
Design Load	5 - HS 20 (Also used for plans with MS 18 and H 20-S16)	
Design Method	LFD	

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COLDWATER CRK LAN

Bridge Inspection Report

Pictures



PHOTO 1 Roadway View

Description Roadway (Coldwater CRK Lane Rd), looking west.



PHOTO 2 Elevation View

Description South bridge elevation view, looking northeast.

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Bridge Inspection Report

Pictures



PHOTO 3 Stream Under View

Description Stream under view, looking southeast.



PHOTO 4 Upstream View

Description Upstream view, looking north.

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Pictures



PHOTO 5 Downstream View

Description Downstream view, looking south.



PHOTO 6 Defect

Description A previously deteriorated area of the east approach concrete pavement next to large concrete patch has been repaired with an asphalt overlay, looking northwest. (Condition Rating: 7)

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Pictures



PHOTO 7 Defect

Description Erosion (up to 6" deep) slightly undermining sidewalk near southwest corner of culvert, looking northeast. (Condition Rating: 6)



PHOTO 8 Defect

Description Vertical and diagonal cracks in southeast wingwall, looking east. (Condition Rating: 7)

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Bridge Inspection Report

Pictures



PHOTO 9 Defect

Description Transverse cracks (< 0.012 " wide) with efflorescence in underside of top slabs, looking south.
(Condition Rating: 7)



PHOTO 10 Defect

Description Large area of erosion behind NE wingwall (up to ~ 2' deep), footing at end of wingwall and upstream apron toewall are exposed up to 1.5' and beginning to undermine, looking south.
(Condition Rating: 6)

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